VECTOR One-Page Model — Homophilic Formation of an Initial League

Last synced: 2026-08-10
Purpose: Define the general dynamic formation model first, then derive the one-shot version used to generate an initial league for the current paper.

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1. Parent Model: Dynamic Bipartite Formation

The general model is a **bipartite affiliation process** between individuals i and teams j.
Each individual has performance/ability $A_{i,t}$. Each team has a lagged centroid:

$$\mu_{j,t-1} = \frac{1}{n_{j,t-1}} \sum_{i \in T_{j,t-1}} A_{i,t-1}.$$

Each team has finite capacity C_j, with remaining openings at assignment step s:

$$R_{j,t}^{(s)} = C_j - n_{j,t}^{(s)}.$$

Define individual–team distance:

$$d_{ij,t} = |A_{i,t}^{*} - \mu_{j,t-1}|.$$

The Quayle-inspired homophily weight is $e^{-\rho d_{ij,t}}$, where ρ controls similarity attraction.
The general assignment rule is:

$$\boxed{P(i \to j \mid t, s) = \frac{R_{j,t}^{(s)} e^{-\rho d_{ij,t}}}{\sum_l R_{l,t}^{(s)} e^{-\rho d_{il,t}}}}$$

for teams with $R_{j,t}^{(s)} > 0$.
After assigning i to j:

$$R_{j,t}^{(s+1)} = R_{j,t}^{(s)} - 1.$$

At the end of period t, new centroids can be calculated and the process repeated:

$$\boxed{\mu_{j,t-1} \to \text{homophilic constrained assignment at } t \to \mu_{j,t} \to \text{assignment at } t+1.}$$

This parent model accommodates later turnover, tenure, reassignment, attrition, and steady-state centroid analysis.

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2. Current Paper: One-Shot Initial-League Specialization

For the present paper, we do **not** need the full longitudinal process.
Use empirical team centroids as fixed initial conditions:

$$\boxed{\mu_j = \mu_{j,0}^{\text{empirical}}.}$$

Use the observed or empirically informed set of individual abilities:

$$\boxed{A_i.}$$

All roster positions are initially open:

$$\boxed{R_j^{(0)} = C \qquad \forall j.}$$

For NCAA men's basketball:

$$\boxed{C = 15.}$$

Randomly permute the players:

$$\pi = (\pi_1, \pi_2, \ldots, \pi_N).$$

At assignment step s, player $i = \pi_s$ chooses among all teams with remaining capacity according to:

$$\boxed{P(i \to j \mid s) = \frac{R_j^{(s)} e^{-\rho|A_i-\mu_j|}}{\sum_l R_l^{(s)} e^{-\rho|A_i-\mu_l|}}}$$

with $R_j^{(s)} > 0$.

When $R_j^{(s)} = 0$, team j automatically has zero probability thereafter.

The algorithm stops when all players are assigned and all roster slots are filled.

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3. Interpretation

At $\rho = 0$:

$$P(i \to j \mid s) = \frac{R_j^{(s)}}{\sum_l R_l^{(s)}}.$$

This is the random capacity-preserving bipartite matching baseline.

As ρ increases, assignments increasingly favor teams whose empirical centroids are close to the player's A_i:

$$\boxed{\text{remaining capacity} \times \text{homophilic similarity attraction.}}$$

The model keeps four concepts separate:

$$\boxed{\mu_j = \text{team state}, \qquad A_i = \text{individual state}, \qquad C = \text{structural constraint}, \qquad \rho = \text{homophilic preference.}}$$

Realized league sorting is an **outcome** of matching, not the same object as ρ.

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4. One Formation Round, Many Realizations

The current model uses **one formation round per generated league**.

We can rerun that same one-shot algorithm many times at fixed ρ:

$$G^{(1)}, G^{(2)}, \ldots, G^{(M)},$$

but these are **independent stochastic realizations**, not longitudinal time steps.

Thus:

$$\boxed{\text{one-shot formation} \neq \text{one stochastic realization only.}}$$

Multiple reruns characterize league-composition variance under the same model conditions without introducing longitudinal dynamics.

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5. Why This Is the Right Baseline

The current ASSIGN mechanism is deliberately minimal:

$$\boxed{\text{empirical team centroids} + \text{individual } A_i + \text{capacity } C + \text{homophily } \rho.}$$

No team prestige term, preferential-attachment term, longitudinal updating, tenure process, or additional sorting parameter is required for the baseline.

Quayle supplies the conceptual precedent: **homophily is a generative preference, while assortativity is measured after formation**. The present model preserves that distinction while replacing one-mode growth with finite-capacity bipartite assignment.

> ***Current paper:*** *generate an initial league from fixed empirical centroids, then repeat independent realizations only as needed to understand variance across generated leagues.*

> ***Later extension:*** *allow $\mu_{j,t}$, roster turnover, tenure, attrition, and reassignment to evolve through time.*